

Antenna Datasheet

Product OC: YEMN117L1B
Version: 1.2
Date: 2026-06-08
Status: Released

Product Name: 4G Screw Mount Low Profile Monopole External Antenna
Key Features:
Frequency Band: 410–470 MHz, 698–2690 MHz
Dimensions: Φ 108 mm × 45 mm
Efficiency: Up to 80 % (MP)
RoHS and REACH Compliant
IP67 & IP69K

Overview

Quectel YEMN117L1B is a 4G puck antenna measuring Φ 108 mm × 45 mm. This ultra-wide-band 4G antenna provides broad coverage from 450–2700 MHz whilst offering backward-compatibility to support 3G and 2G networks as well as LTE Cat-M and narrowband IoT (NB-IoT). The antenna is available with connection via cable lengths from 300–5,000 mm, terminated with SMA Male connector. Ideal for applications where the antenna is required to be discrete, this low profile, screw mount omni-directional antenna is easy to install with maximum durability assured thanks to its IP67 & IP69K and IK10 rated, PC enclosure. It is compatible with Quectel's RM520x Series modules.

It allows constant and reliable transmission and reception due to its omni-directional gain across all frequency bands. YEMN117L1B is designed as a monopole antenna, which offers high efficiency in all working bands. It is a perfect antenna product for customers that desire highest performance. This high-efficiency, high-gain omni-directional antenna is ideally suited for smart metering, remote monitoring, vehicle tracking and telematics, and many other IoT devices. It is suitable for outdoor and indoor applications due to its robust UV resistant and flame-resistant PC enclosure meets UL 746c f1 and UL 94 V-0.

- **Typical applications include:**

✓ High speed video
✓ Real-time streaming
✓ High-capacity MIMO networks
✓ Public transportation

Quectel provides comprehensive antenna design support such as simulation, testing and manufacturing for custom antenna solutions to meet your specific application needs. We have regional R & D centers to offer quick response to meet your requirements. Please contact our sales & FAEs if you have any requests.

Contents

Overview **1**
Contents **2**

1 Specification **3**
1.1. Electrical 3
1.2. Mechanical & Environmental 6

2 Drawing **7**

3 Detailed Performance **8**
3.1. S-Parameter Test 8
3.1.1. VSWR 8
3.1.2. Return Loss 10
3.2. Radiation Performance Test 12
3.2.1. Efficiency 12
3.2.2. Average Gain 14
3.2.3. Peak Gain 16
3.2.4. 3D & 2D Radiation Pattern 18
3.2.4.1. Test Condition: Free Space 18
3.2.4.2. Test Condition: On 300 mm × 300 mm Metal Plane 22

4 Packaging **26**

Contact Us **28**
Legal Notices **29**
Revision History **31**

1 Specification

Test Condition: In Free Space & On 300 mm × 300 mm Metal Plane

1.1. Electrical

Electrical	
Frequency Range	410–470 MHz, 698–2690 MHz
Impedance	50 Ω
Polarization	Linear
Radiation Pattern	Omni-directional

Electrical – Detail

Band / SPEC	Band	B71	B12 /B13 /B28	B5 /B8 /B26	n74 /n75 /n76	B1 /B2 /B3	B40	Wi-Fi 2G	B38 /B41	B42 /B48 /n77	n79	Wi-Fi 5G
	Freq. (MHz)	600–700	700–810	820–960	1420–1520	1700–2170	2300–2400	2400–2500	2500–2690	3300–4200	4400–5000	5150–5850
Max. VSWR	FS	1.8	1.8	3.6	2.7	2.5	2.4	1.9	1.8	-	-	-
	MP	1.8	4.2	5.0	2.8	2.8	2.6	2.2	1.7	-	-	-
Max. Return Loss (dB)	FS	-10.8	-10.6	-5.0	-6.7	-7.3	-7.8	-10.3	-11.1	-	-	-
	MP	-11.1	-4.2	-3.6	-6.5	-6.5	-7.0	-8.3	-11.4	-	-	-
AVG Eff. (%)	FS	60.7	64.9	55.2	56.8	48.4	47.3	49.9	51.9	-	-	-
	MP	77.0	64.5	39.6	47.8	49.0	47.7	51.5	52.6	-	-	-
AVG. AVG Gain (dB)	FS	-2.2	-1.9	-2.6	-2.5	-3.2	-3.2	-3.0	-2.8	-	-	-
	MP	-1.1	-1.9	-4.0	-3.2	-3.1	-3.2	-2.9	-2.8	-	-	-
Max. Peak Gain (dBi)	FS	2.8	4.6	4.1	4.7	4.9	3.9	3.5	4.8	-	-	-
	MP	6.0	6.3	4.4	6.9	6.8	6.2	6.2	6.2	-	-	-

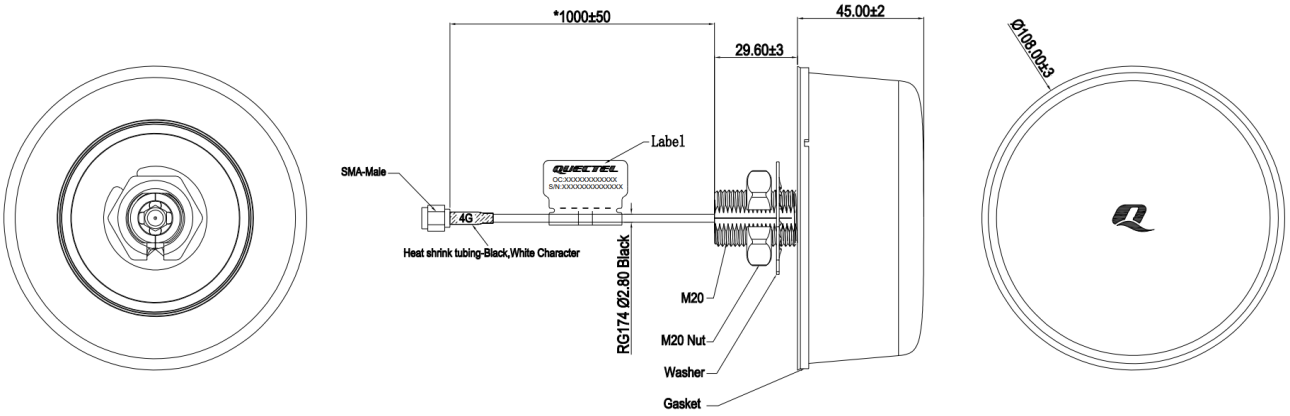
VSWR	FS	≤ 3.6
	MP	≤ 5.0
Return Loss	FS	≤ -5.0 dB
	MP	≤ -3.6 dB
Peak Gain	FS	≤ 4.9 dBi
	MP	≤ 6.9 dBi

Electrical – Detail			
Band / SPEC	Band	B87/B88	B31/B72/B73
	Freq. (MHz)	410–430 MHz	450–470 MHz
Max. VSWR	FS	2.3	1.7
	MP	2.4	1.7
Max. Return Loss (dB)	FS	-8.1	-11.9
	MP	-7.5	-11.8
AVG Eff. (%)	FS	39.1	45.0
	MP	37.0	42.8
AVG. AVG Gain (dB)	FS	-4.1	-3.5
	MP	-4.3	-3.7
Max. Peak Gain (dBi)	FS	1.4	2.0
	MP	4.0	4.7
VSWR	FS	≤ 2.3	
	MP	≤ 2.4	
Return Loss	FS	≤ -8.1 dB	
	MP	≤ -7.5 dB	
Peak Gain	FS	≤ 2.0 dBi	
	MP	≤ 4.7 dBi	

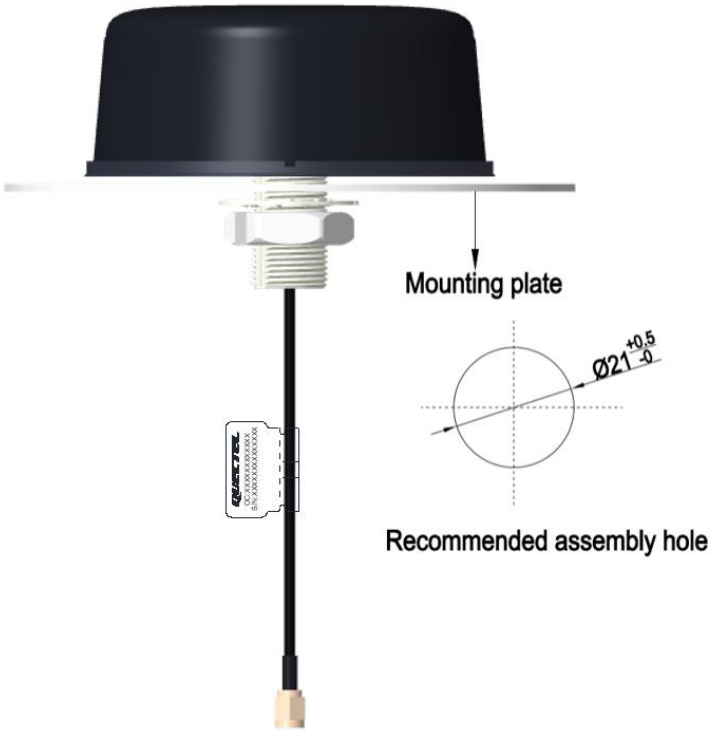
1.2. Mechanical & Environmental

Mechanical	
Antenna Dimensions	Φ 108 mm × 45 mm
Material & Color	PC & Black
Connector Type	SMA Male (The current state of the SMA connector is not waterproof. If a waterproof connector is required, it can be customized.)
Mounting Type	Screw Mounting (M20)
Weight	Typ. 260 g
Environmental	
Operation Temperature	-40 °C to +85 °C
Storage Temperature	-40 °C to +85 °C
Ingress Protection (IP) Rating	IP67 (After Installation) IP69K (After Installation)
Impact Protection (IK) Rating	IK10 (After Installation)
RoHS & REACH Compliant	Yes
Housing Flame Rating	UL 94 V-0
Housing UV Resistant	UL 746c f1

2 Drawing



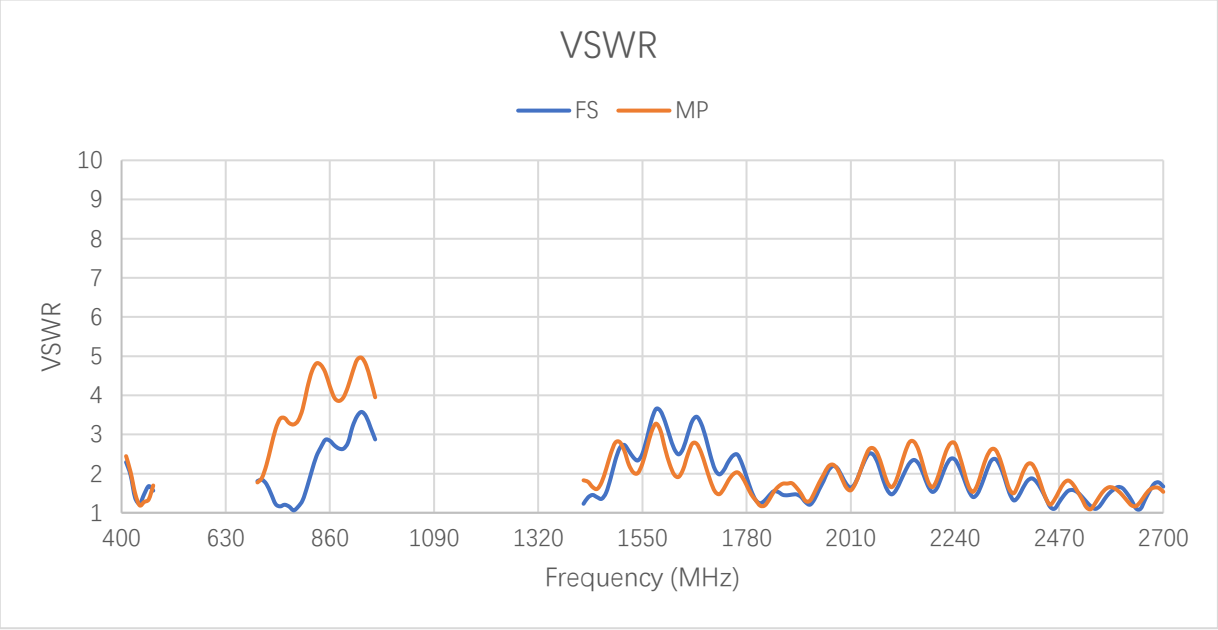
- Recommended hole dimensions as below view.
- Recommended mounting plate thickness: 2–10 mm.
- Recommended mounting plate size: ≥ φ 85 mm



3 Detailed Performance

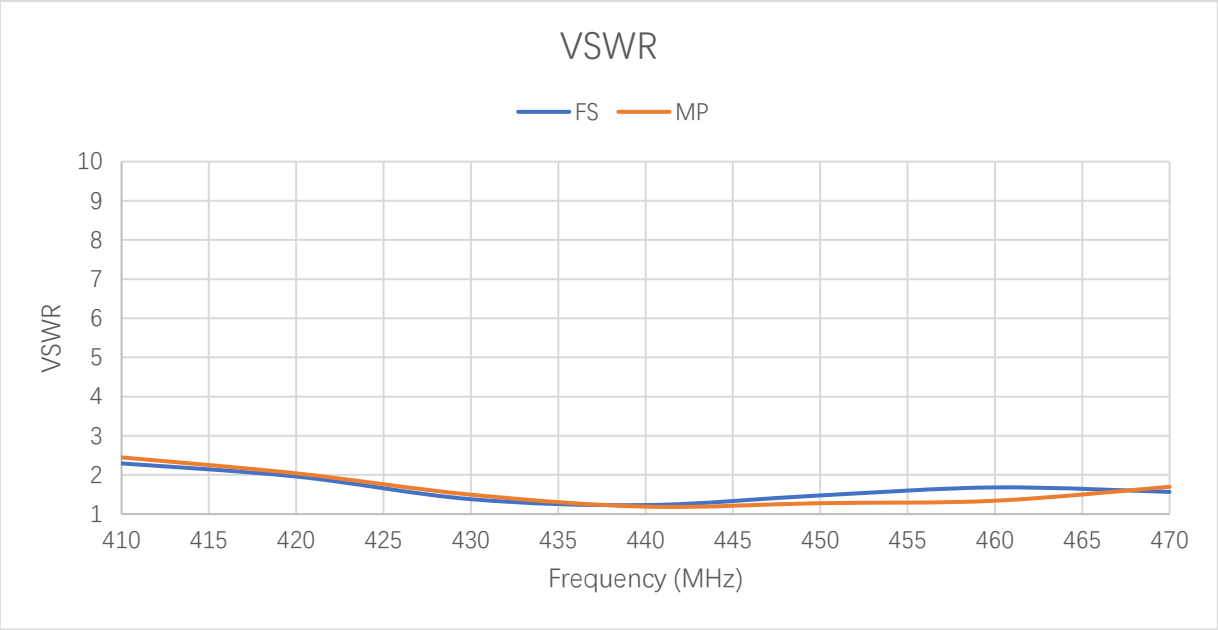
3.1. S-Parameter Test

3.1.1. VSWR



VSWR

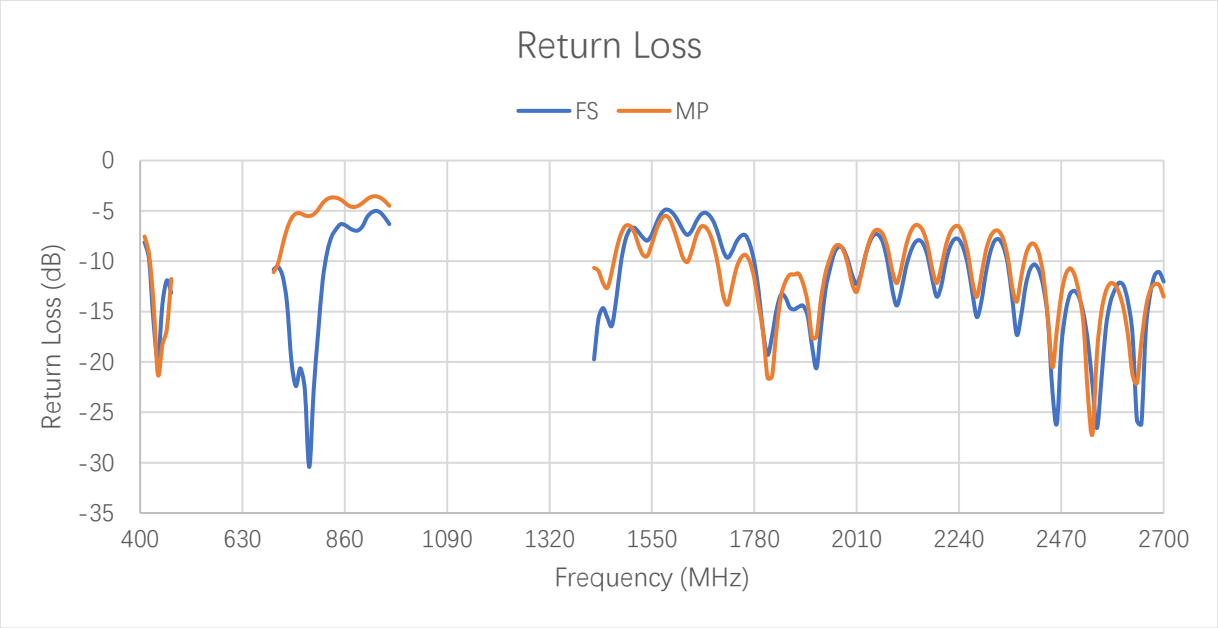
Frequency (MHz)		410	450	710	830	900	960	1440	1710	1740	1880
VSWR	FS	2.3	1.5	1.8	2.4	2.8	2.9	1.5	2.1	2.3	1.5
	MP	2.4	1.3	1.9	4.8	4.2	3.9	1.7	1.5	1.8	1.8
Frequency (MHz)		**1950**	**2140**	**2350**	**2450**	**2600**	**2690**	**4700**	**5000**	**5500**	**6000**
VSWR	FS	1.8	2.3	1.9	1.2	1.7	1.8	-	-	-	-
	MP	2.0	2.8	2.0	1.2	1.6	2.0	-	-	-	-



VSWR

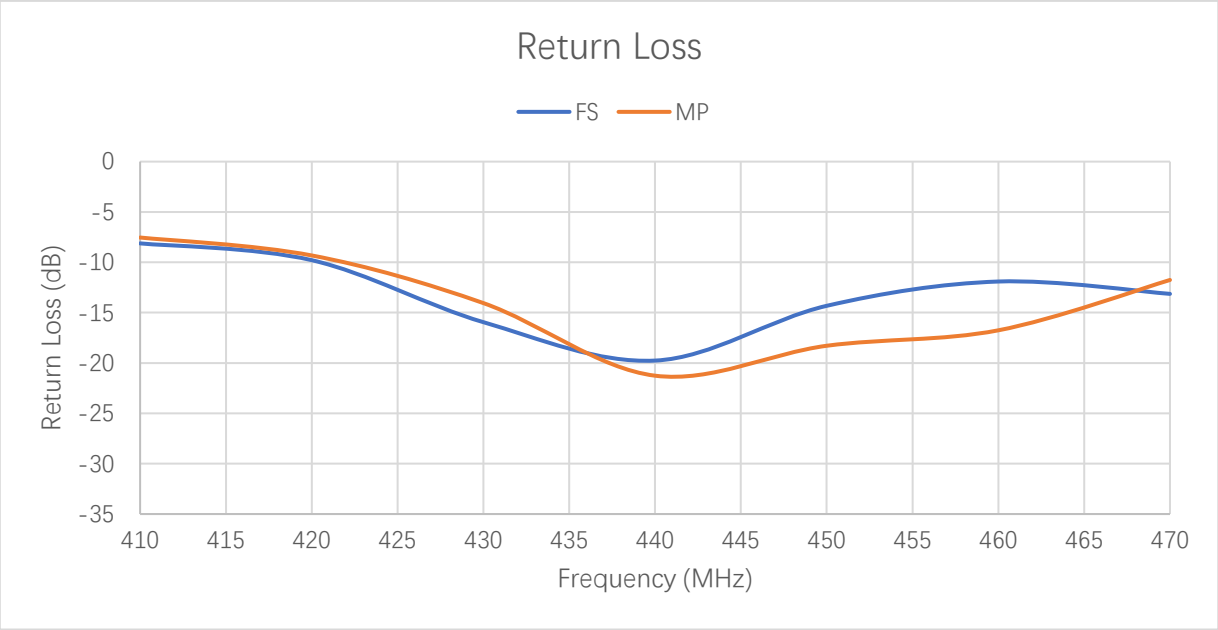
Frequency (MHz)	410	420	430	440	450	460	470
FS	2.3	2.0	1.4	1.2	1.5	1.7	1.6
MP	2.4	2.0	1.5	1.2	1.3	1.3	1.7

3.1.2. Return Loss



Return Loss (dB)

Frequency (MHz)		410	450	710	830	900	960	1440	1710	1740	1880
Return Loss (dB)	FS	-8.1	-14.3	-10.6	-7.6	-6.5	-6.3	-14.6	-8.9	-8.1	-14.5
	MP	-7.5	-18.3	-10.2	-3.7	-4.2	-4.5	-12.1	-13.4	-10.7	-11.2
Frequency (MHz)		**1950**	**2140**	**2350**	**2450**	**2600**	**2690**	**4700**	**5000**	**5500**	**6000**
Return Loss (dB)	FS	-10.8	-8.2	-10.3	-22.9	-12.1	-10.8	-	-	-	-
	MP	-9.7	-6.5	-9.7	-20.5	-13.2	-9.7	-	-	-	-

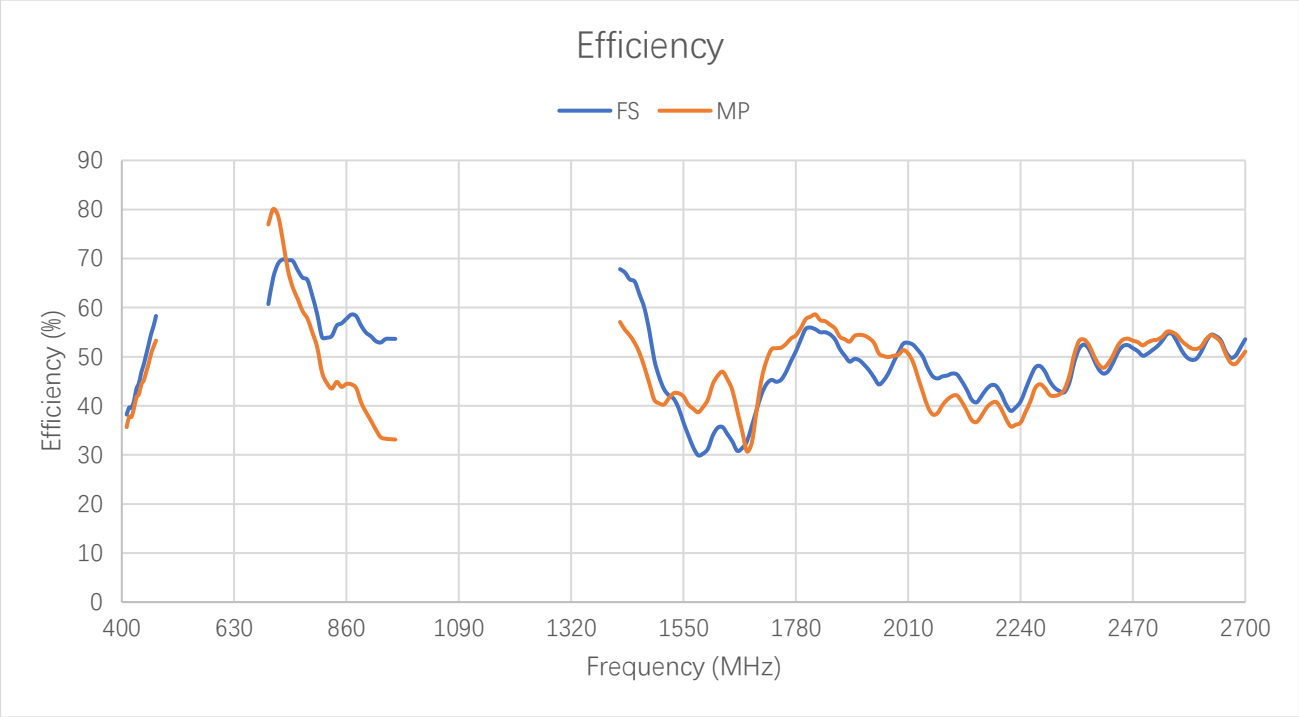


Return Loss (dB)

Frequency (MHz)	410	420	430	440	450	460	470
FS	-8.1	-9.8	-15.9	-19.8	-14.3	-11.9	-13.1
MP	-7.5	-9.3	-14.1	-21.3	-18.3	-16.8	-11.8

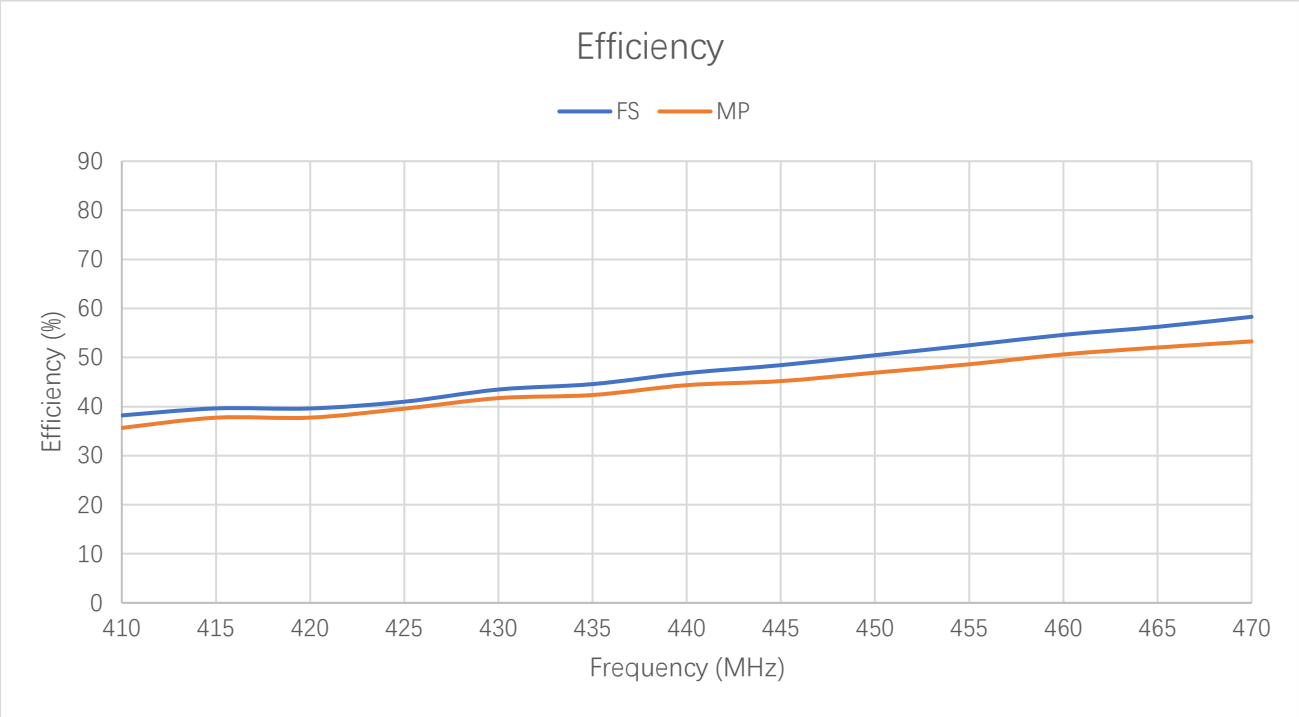
3.2. Radiation Performance Test

3.2.1. Efficiency



Efficiency (%)

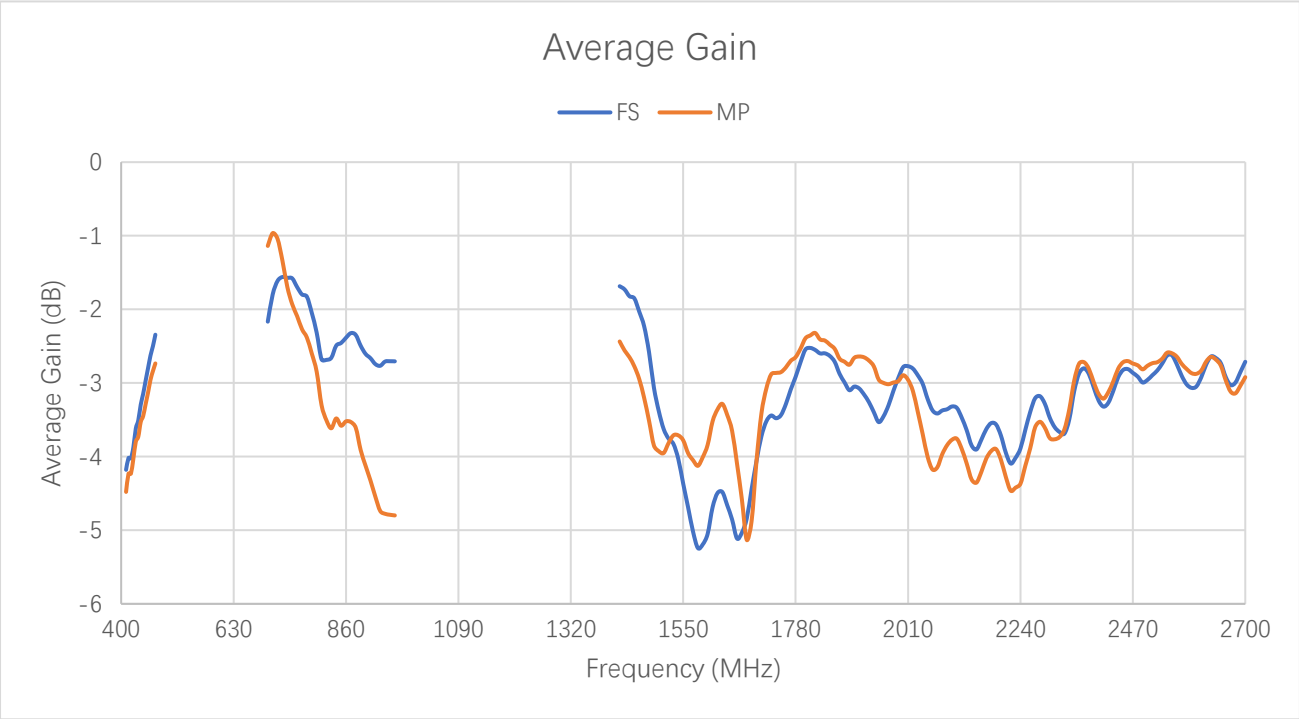
Frequency (MHz)		410	450	710	830	900	960	1440	1710	1740	1880
Efficiency (%)	FS	38.2	50.5	66.1	54.2	54.9	53.6	65.3	42.4	44.9	50.2
	MP	35.7	46.9	80.1	43.5	38.7	33.1	52.8	45.6	51.7	53.6
Frequency (MHz)		**1950**	**2140**	**2350**	**2450**	**2600**	**2690**	**4700**	**5000**	**5500**	**6000**
Efficiency (%)	FS	44.4	41.2	48.9	52.2	49.6	44.4	-	-	-	-
	MP	50.7	37.1	50.5	53.5	51.6	50.7	-	-	-	-



Efficiency (%)

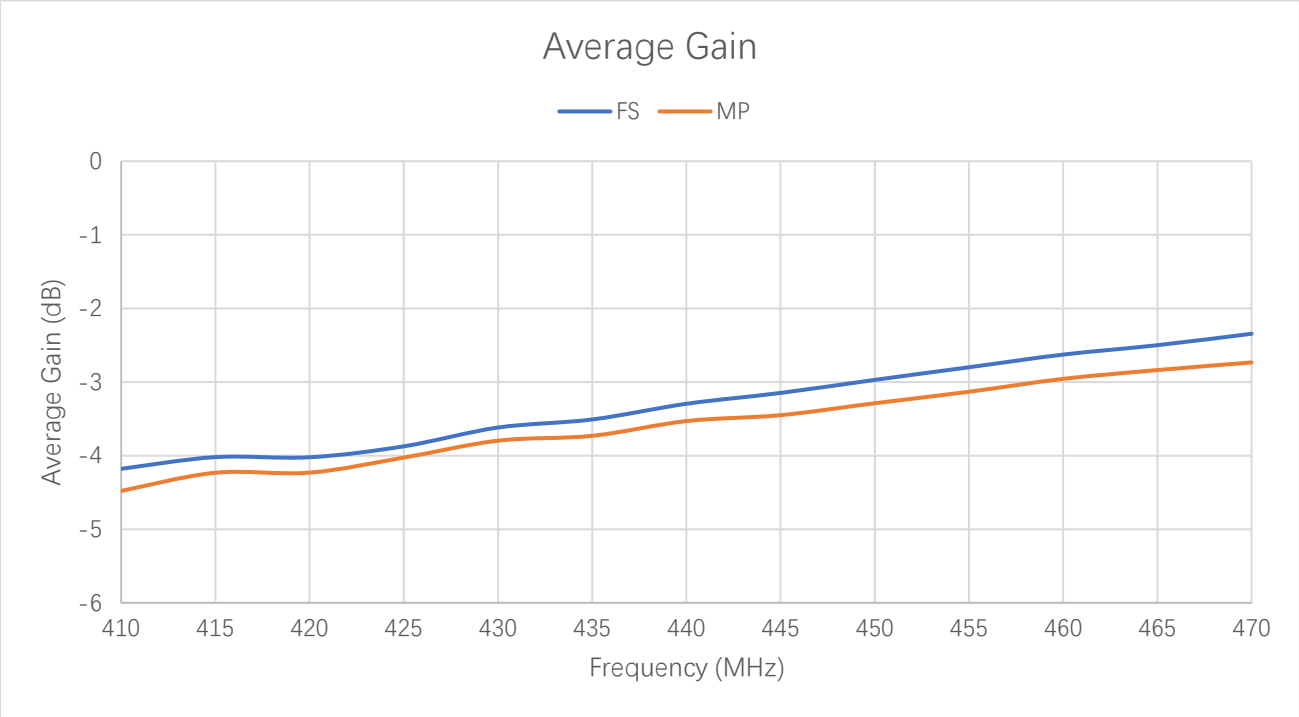
Frequency (MHz)	410	420	430	440	450	460	470
FS	38.2	39.6	43.5	46.8	50.5	54.6	58.3
MP	35.7	37.8	41.7	44.4	46.9	50.6	53.3

3.2.2. Average Gain



Average Gain (dB)

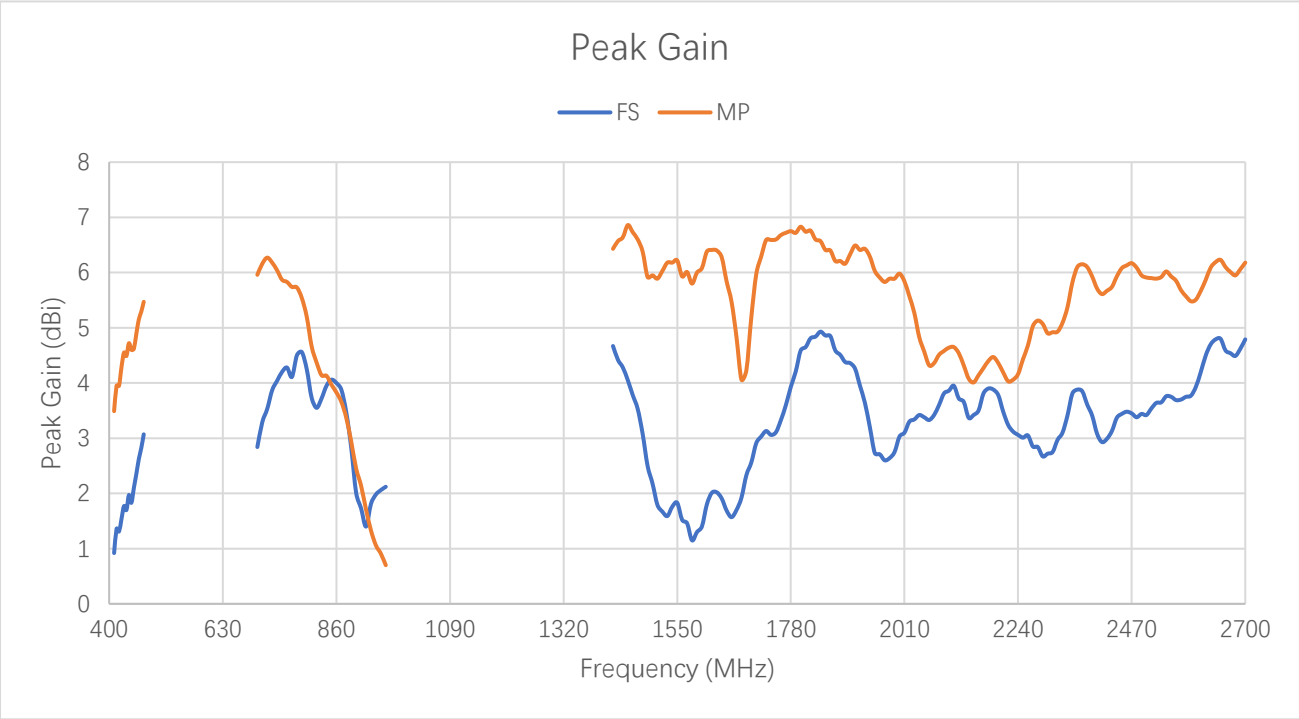
Frequency (MHz)		410	450	710	830	900	960	1440	1710	1740	1880
Average Gain (dB)	FS	-4.2	-3.0	-1.8	-2.7	-2.6	-2.7	-1.9	-3.7	-3.5	-3.0
	MP	-4.5	-3.3	-1.0	-3.6	-4.1	-4.8	-2.8	-3.4	-2.9	-2.7
Frequency (MHz)		**1950**	**2140**	**2350**	**2450**	**2600**	**2690**	**4700**	**5000**	**5500**	**6000**
Average Gain (dB)	FS	-3.5	-3.8	-3.1	-2.8	-3.0	-3.5	-	-	-	-
	MP	-3.0	-4.3	-3.0	-2.7	-2.9	-3.0	-	-	-	-



Average Gain (dB)

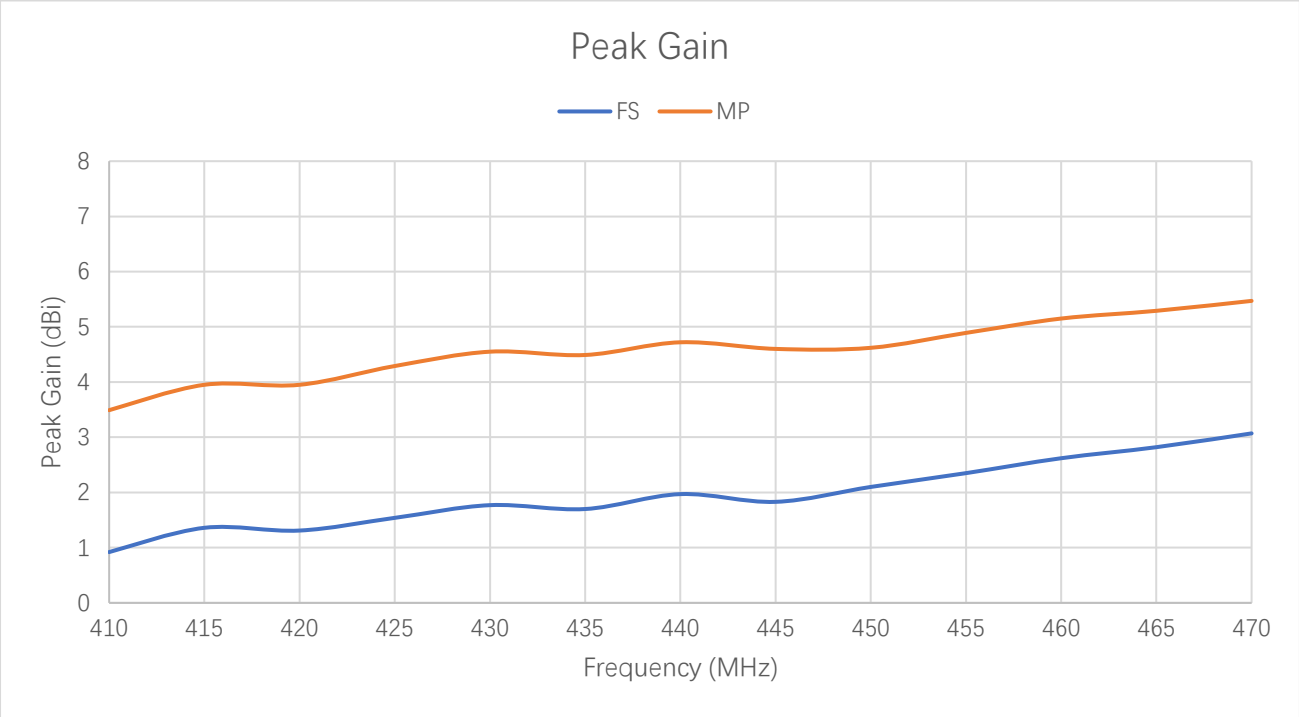
Frequency (MHz)	410	420	430	440	450	460	470
FS	-4.2	-4.0	-3.6	-3.3	-3.0	-2.6	-2.3
MP	-4.5	-4.2	-3.8	-3.5	-3.3	-3.0	-2.7

3.2.3. Peak Gain



Peak Gain (dBi)

Frequency (MHz)		410	450	710	830	900	960	1440	1710	1740	1880
Peak Gain (dBi)	FS	0.9	2.1	3.3	3.7	2.0	2.1	4.1	2.9	3.1	4.5
	MP	3.5	4.6	6.2	4.1	2.5	0.7	6.9	6.0	6.6	6.2
Frequency (MHz)		**1950**	**2140**	**2350**	**2450**	**2600**	**2690**	**4700**	**5000**	**5500**	**6000**
Peak Gain (dBi)	FS	2.7	3.4	3.8	3.4	3.9	2.7	-	-	-	-
	MP	6.0	4.1	5.8	6.1	5.5	6.0	-	-	-	-



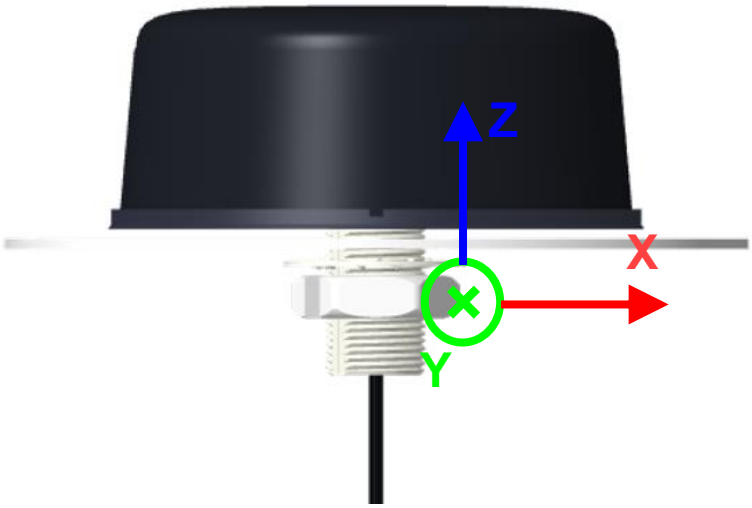
Peak Gain (dBi)

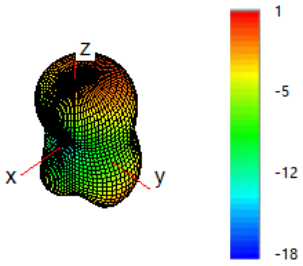
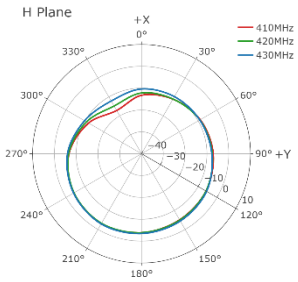
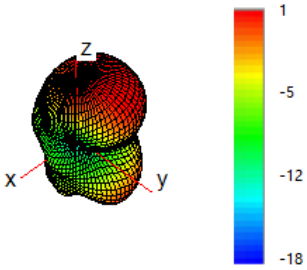
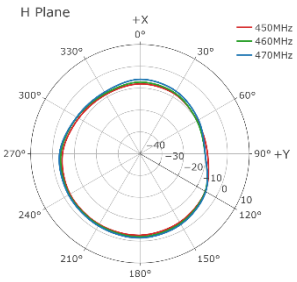
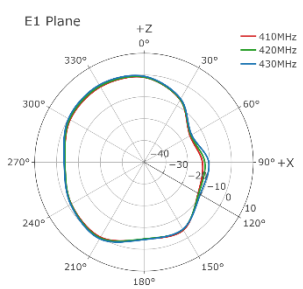
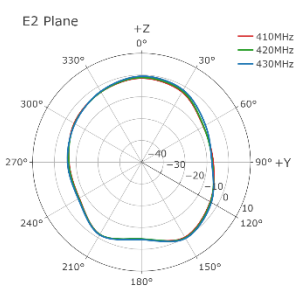
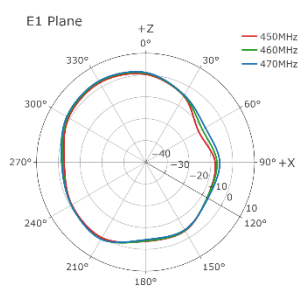
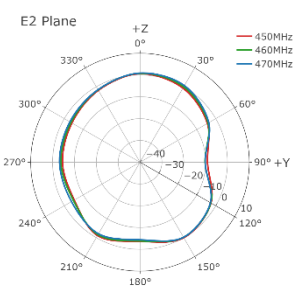
Frequency (MHz)	410	420	430	440	450	460	470
FS	0.9	1.3	1.8	2.0	2.1	2.6	3.1
MP	3.5	4.0	4.6	4.7	4.6	5.2	5.5

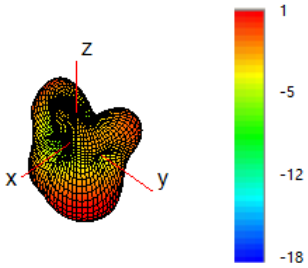
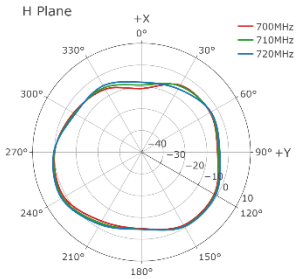
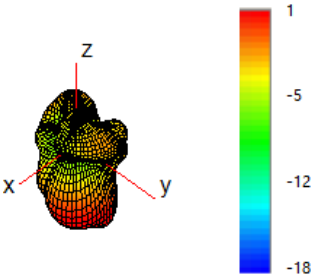
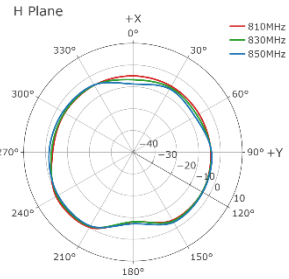
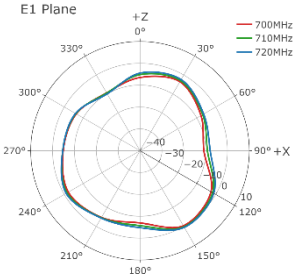
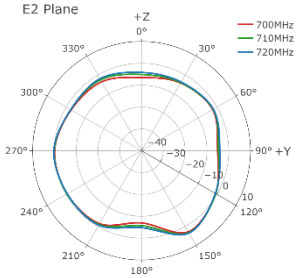
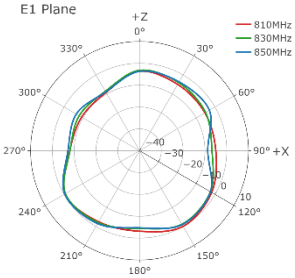
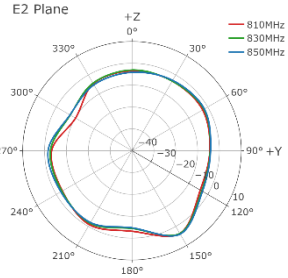
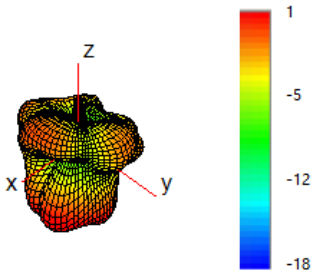
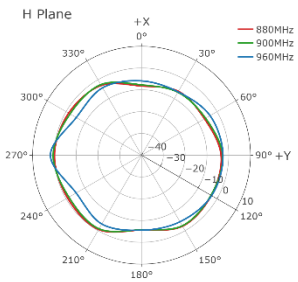
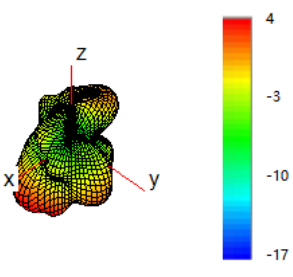
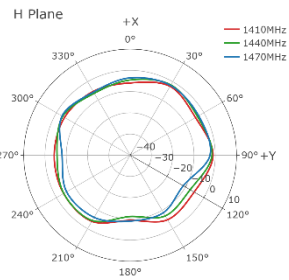
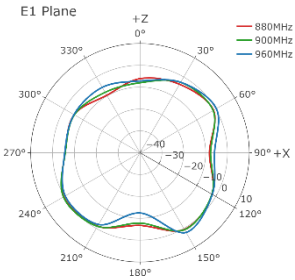
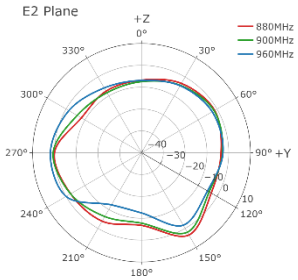
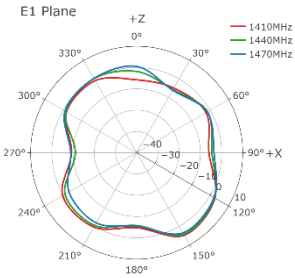
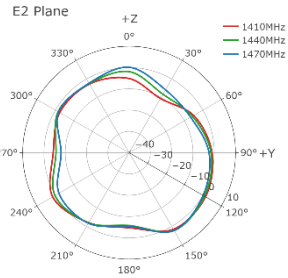
3.2.4. 3D & 2D Radiation Pattern

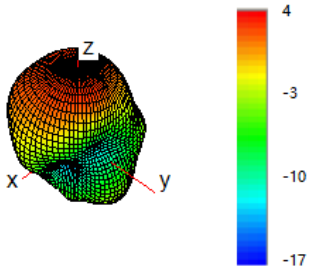
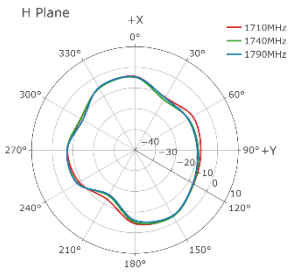
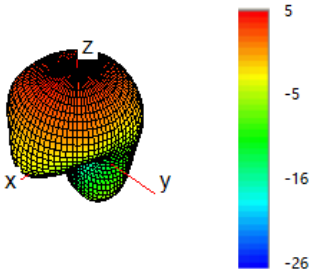
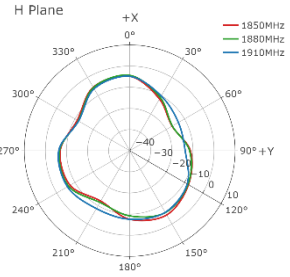
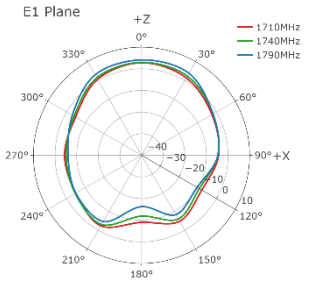
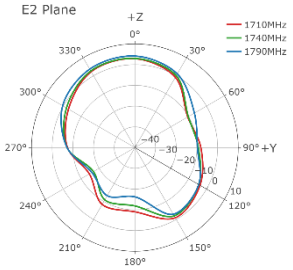
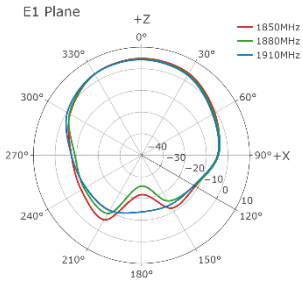
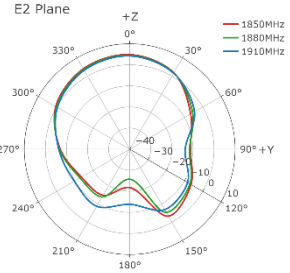
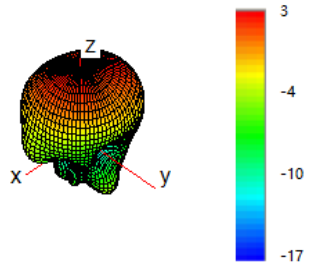
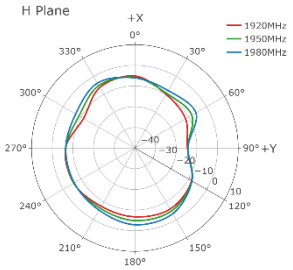
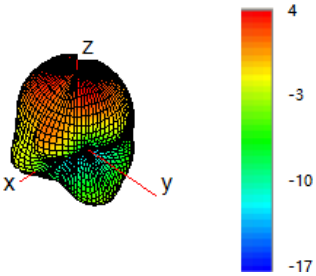
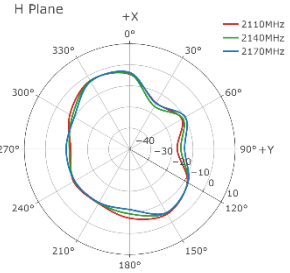
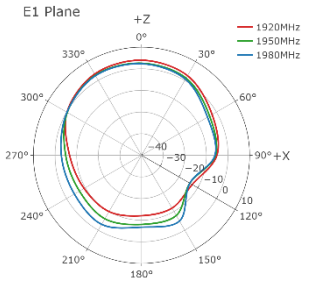
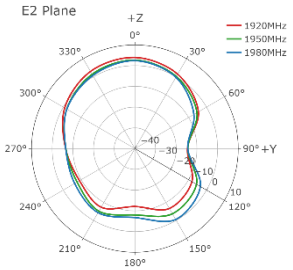
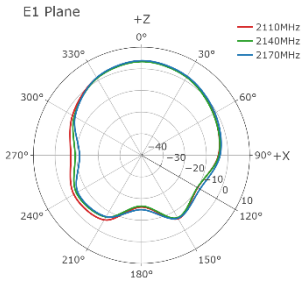
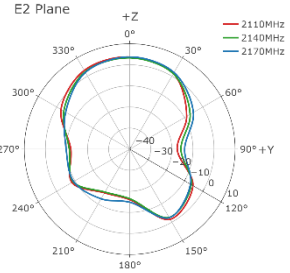
3.2.4.1. Test Condition: Free Space

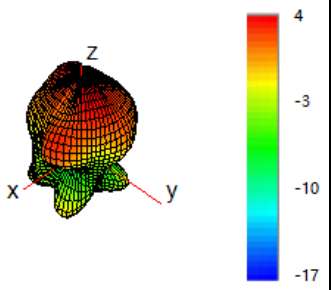
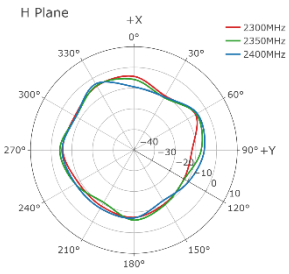
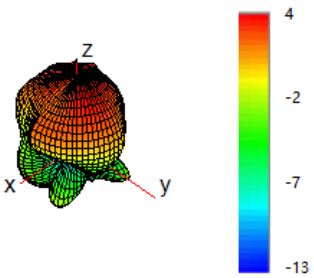
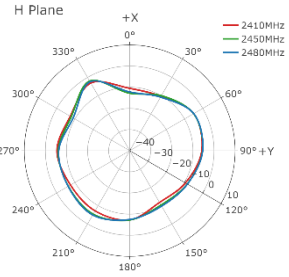
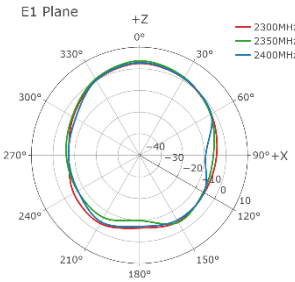
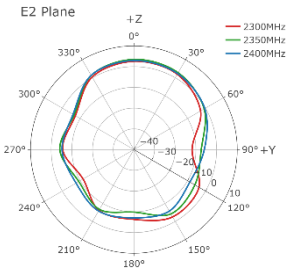
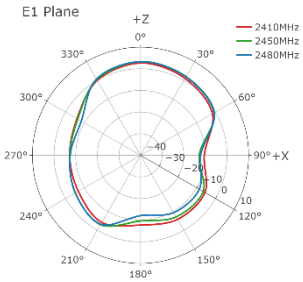
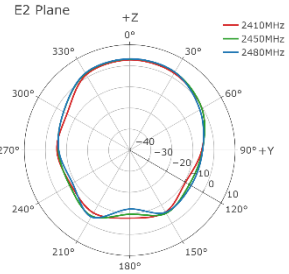
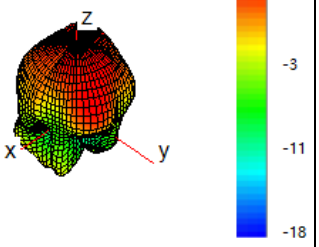
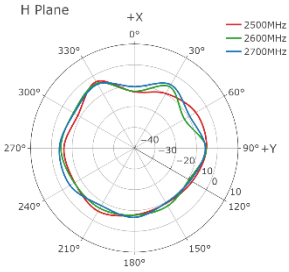
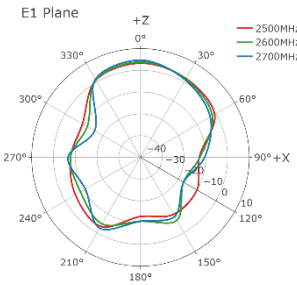
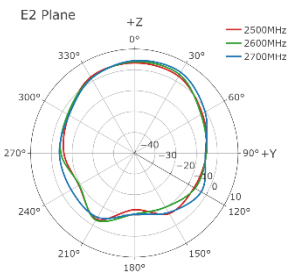
- Test Chamber: FS-G-1



420 MHz		460 MHz	
			
			

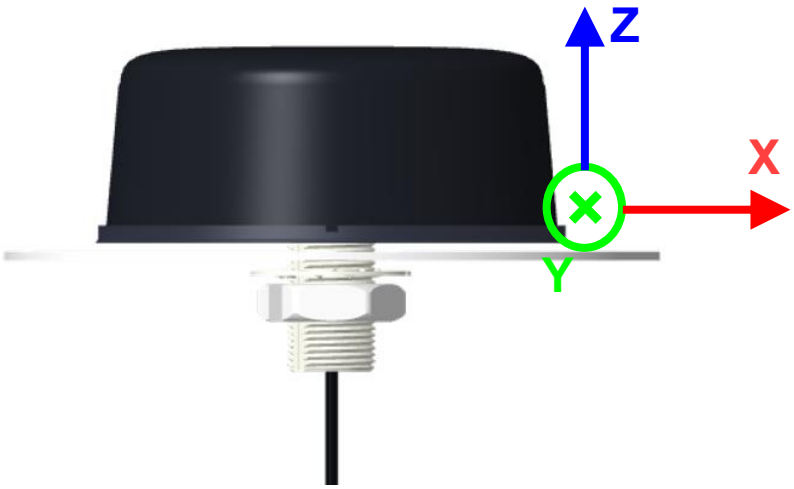
710 MHz		830 MHz	
			
			
900 MHz		1440 MHz	
			
			

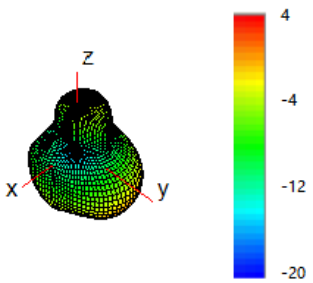
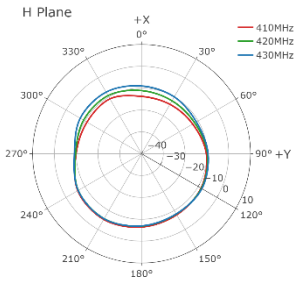
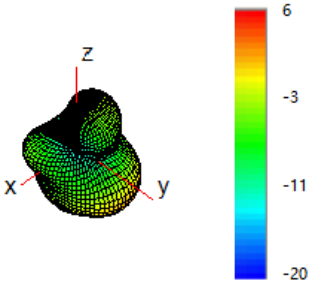
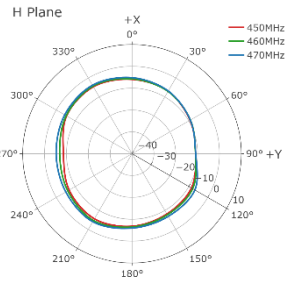
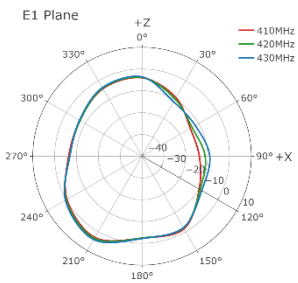
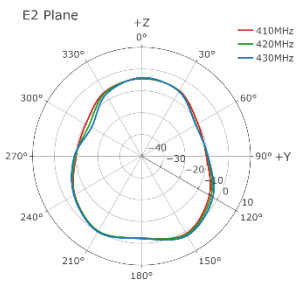
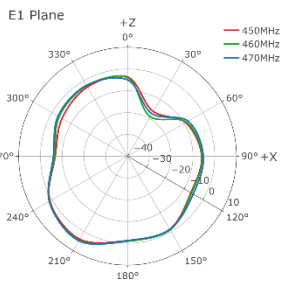
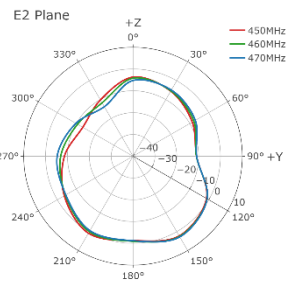
1740 MHz		1880 MHz	
			
			
1950 MHz		2140 MHz	
			
			

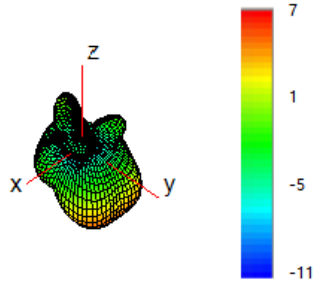
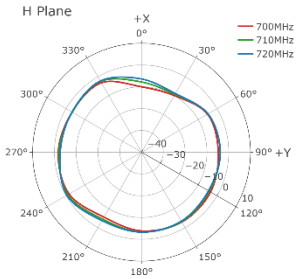
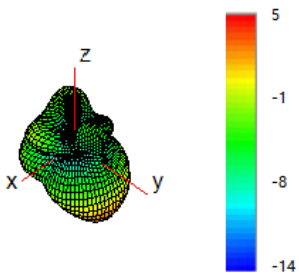
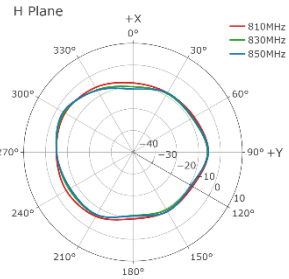
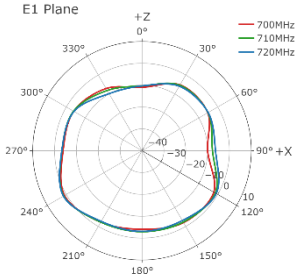
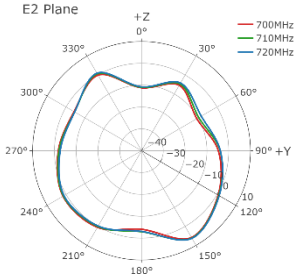
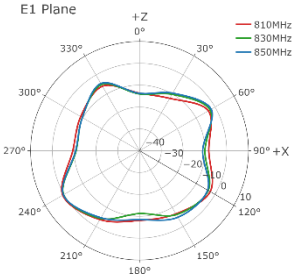
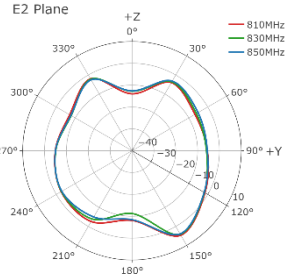
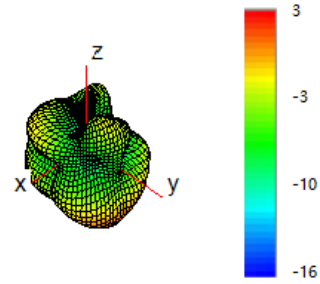
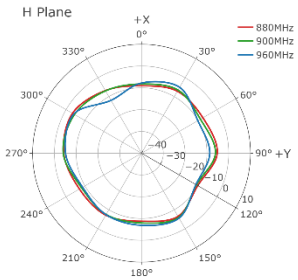
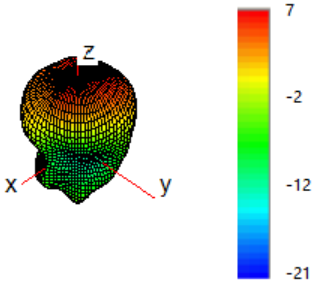
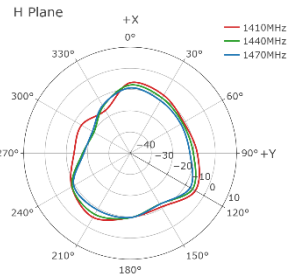
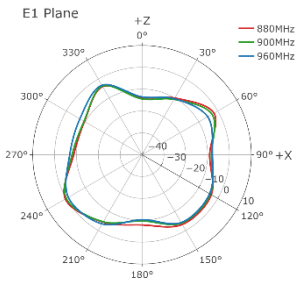
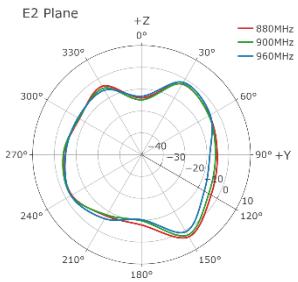
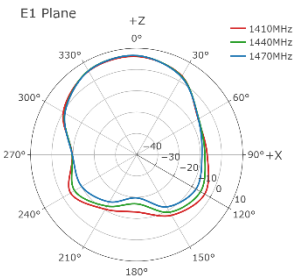
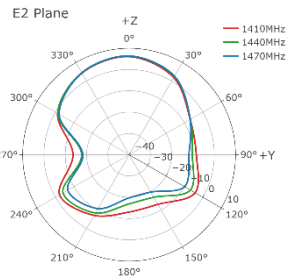
2350 MHz		2450 MHz	
			
			
2600 MHz			
			
			

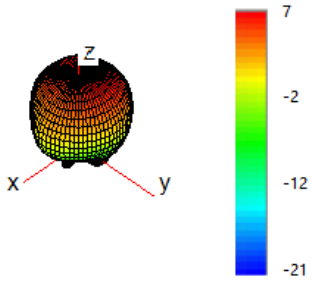
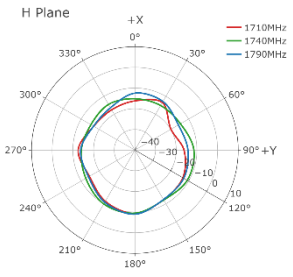
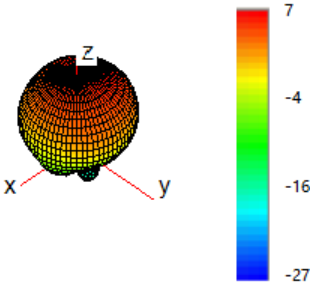
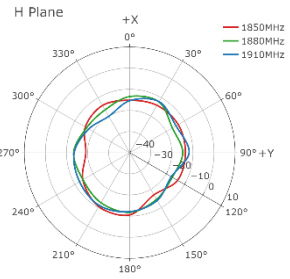
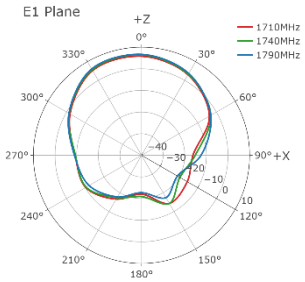
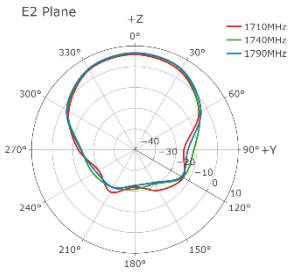
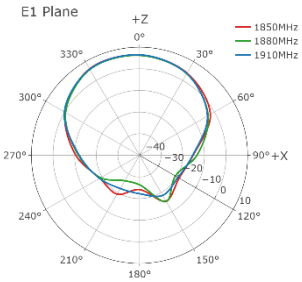
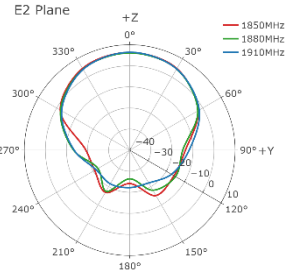
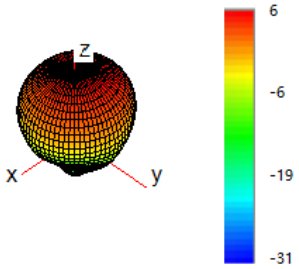
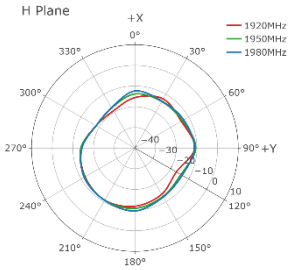
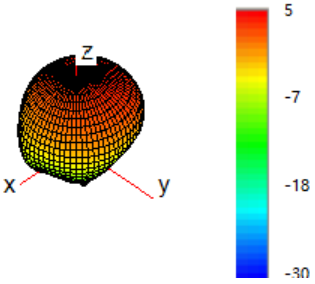
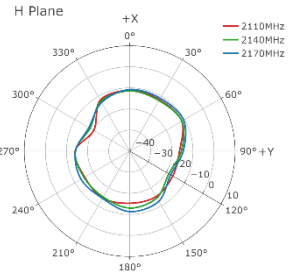
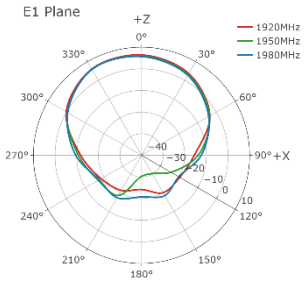
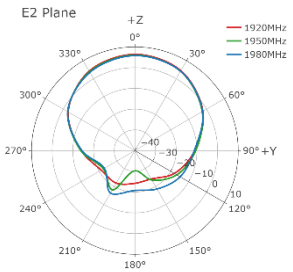
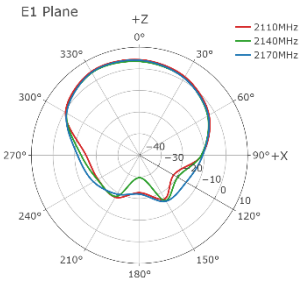
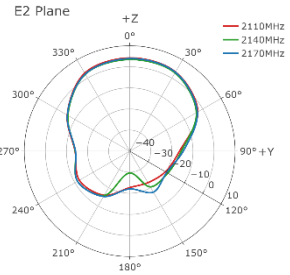
3.2.4.2. Test Condition: On 300 mm × 300 mm Metal Plane

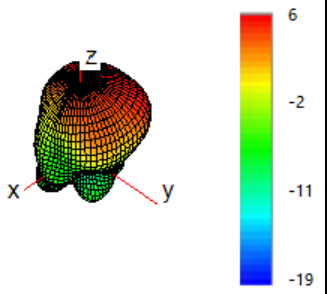
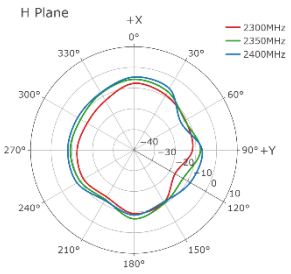
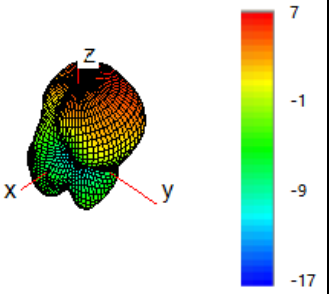
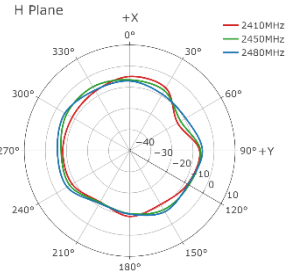
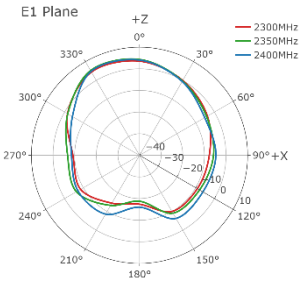
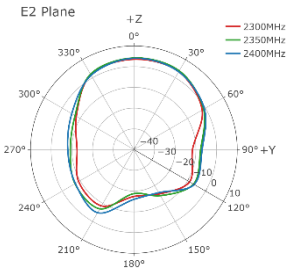
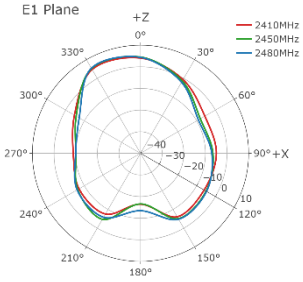
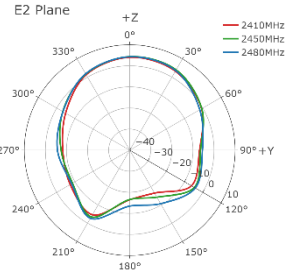
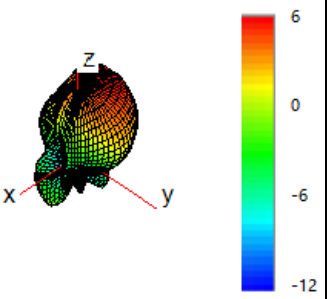
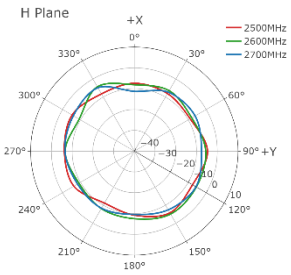
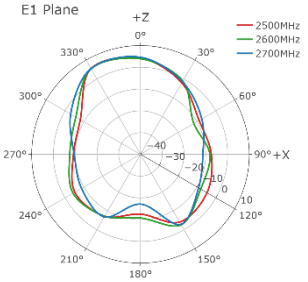
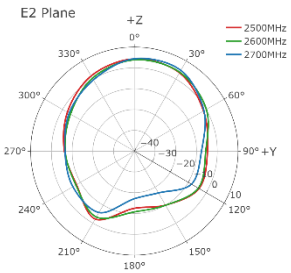
- Test Chamber: FS-G-1



420 MHz		460 MHz	
			
			

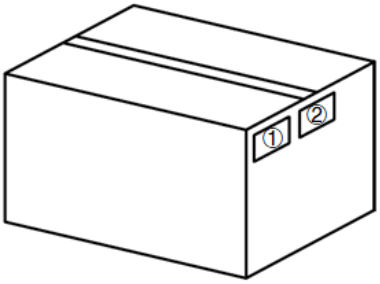
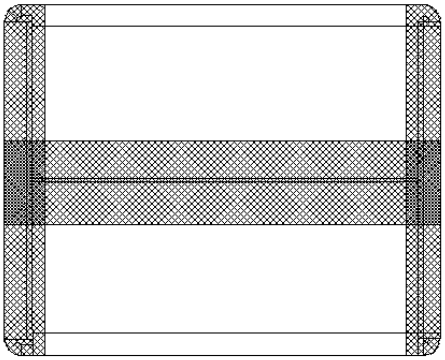
710 MHz		830 MHz	
			
			
900 MHz		1440 MHz	
			
			

1740 MHz		1880 MHz	
			
			
1950 MHz		2140 MHz	
			
			

2350 MHz		2450 MHz	
			
			
2600 MHz			
			
			

4 Packaging

Step	Packaging Picture / 2D Picture	Description
1		Wrap the product in a PE bag and put it into the inner box.
2		Place the pearl cotton on the product.
3		Inner box diagram Inner Box Size: L × W × H = 165 × 165 × 135 mm

4		(18 Inner Boxes / Carton Box) (18 Antennas / Carton Box) Carton Size: L × W × H = 525 × 525 × 305 mm
5		**Position for Attaching Labels** ① Carton Label ② Quality Label
6		**Sealing Cartons** H-shaped sealing cartons

Contact Us

At Quectel, our aim is to provide timely and comprehensive services to our customers. If you require any assistance, please contact our headquarters:

Quectel Wireless Solutions Co., Ltd.
No. 8 Waipojing Road, Sijing Town, Songjiang District, Shanghai 201601, China
Tel: +86 21 5108 6236
Email: info@quectel.com

Or our local offices. For more information, please visit:
https://www.quectel.com/contact/.

For technical support, or to report documentation errors, please visit:
https://www.quectel.com/tech-support/.
Or email us at: support@quectel.com.

Legal Notices

We provide this document to support your product design. You are required to design your products based on the specifications and parameters set forth herein. You agree that you are responsible for using independent analysis and evaluation in designing intended products, and we provide reference designs for illustrative purposes only. Before using any hardware, software or service guided by this document, please read this notice carefully. Even though we employ commercially reasonable efforts to provide the best possible experience, you hereby acknowledge and agree that this document and related services hereunder are provided to you on an “as available” basis. You acknowledge and agree that we may add to, amend, or restate this document at any time at our sole discretion without any prior notice to you, and such additions, amendments, or restatements shall be binding upon you.

Use and Disclosure Restrictions

License Agreements

The recipient of any hardware, software, materials, or documentation provided by us shall keep such content confidential, unless expressly authorized by us. The recipient shall not disclose, access, or use any part of the received content for any purpose other than the execution and implementation of the intended project.

Copyright

Our and third-party products hereunder may contain copyrighted materials, including but not limited to protected content, hardware, software, and documentation owned by us or applicable third parties. Unless prior written consent is obtained, you shall not access, use, or disclose any documents or information provided by us, nor shall you copy, reproduce, republish, display, translate, distribute, merge, modify, or create derivative works from any such copyrighted materials. We and the applicable third party retain exclusive rights to all copyrighted materials. No license to any patents, copyrights, trademarks, or service marks shall be granted or transferred. For the avoidance of doubt, no form of purchase shall be construed as granting any license beyond a normal, non-exclusive, royalty-free license to use the product. We reserve the right to pursue legal action against any violation of confidentiality obligations, unauthorized use, or any other unlawful or malicious use of the aforementioned documents and information.

Trademarks

Unless otherwise expressly provided, nothing in this document shall be construed as conferring any rights to use any trademark, trade name, name, abbreviation, or counterfeit thereof owned by us or any third party in advertising, publicity, or any other contexts.

Third-Party Rights

You understand that this document may refer to hardware, software, and/or documentation owned by one or more third parties (“third-party materials”). Use of such third-party materials is subject to all applicable restrictions and obligations set forth herein.

We make no warranty or representation, either express or implied, regarding the third-party materials, including but not limited to any implied or statutory, warranties of merchantability or fitness for a particular purpose, quiet enjoyment, system integration, information accuracy, and non-infringement of any third-party intellectual property rights with regard to the licensed technology or use thereof. Nothing herein constitutes a representation or warranty by us to either develop, enhance, modify, distribute, market, sell, offer for sale, or otherwise maintain production of any our products or any other hardware, software, device, tool, information, or product. We moreover disclaim any and all warranties arising from the course of dealing, course of performance, or usage of trade.

Privacy Policy

To enable product functionality, certain device data may be uploaded to our or third-party servers, including those operated by carriers, chipset suppliers, or servers designated by you. We strictly comply with applicable laws and regulations and will retain, use, disclose, or otherwise process relevant data solely for the purpose of enabling product functionality, or as permitted by applicable laws. Before interacting with any third party regarding data exchange, please be informed of and understand their privacy and data security policies.

Disclaimer

a) We shall not be liable for any damages resulting from failure to comply with applicable operational or design specifications.
b) We shall bear no liability for any inaccuracies or omissions in this document, nor for any damages arising from the use of the information contained herein.
c) While we make every effort to ensure the integrity, accuracy, and timeliness of the features and functions under development, errors or omissions may nevertheless occur. Unless otherwise provided in a valid written agreement, we make no warranties of any kind, express, implied, or statutory, and disclaim all liability for any loss or damage arising from the use of any features or functions under development, to the maximum extent permitted by law, regardless of whether such loss or damage is foreseeable.
d) We assume no legal responsibility for the accessibility, safety, accuracy, availability, legality, or completeness of any information, content, advertising, commercial offers, products, services, or materials on third-party websites or third-party resources.

Copyright © Quectel Wireless Solutions Co., Ltd. 2026. All rights reserved.

Revision History

Version	Date	Author	Note
-	2024-03-19	Mordecai Liu/ Lance Sun/ David Liu/ Rainey Liao	Creation of the document
1.0	2024-03-19	Mordecai Liu/ Lance Sun/ David Liu/ Rainey Liao	First official release
1.1	2025-06-20	Mordecai Liu/ Strong Qiang/ Rainey Liao	1. Updated the supported antenna frequency band to 698–2690 MHz. 2. Updated the antenna image (Cover page). 3. Listed the related data of 410-470 MHz separately (Chapter 3). 4. Updated the packaging (Chapter 4).
1.2	2026-06-08	Blake Xiang	Updated the IP rating (Chapter 1.2).



www.quectel.com